

Practical 6

Develop application to download image/video from server or upload image/video to server using MTOM techniques

Software Tools Required

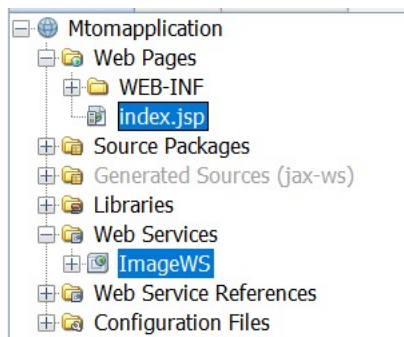
- **IDE or Code Editor:** Netbeans version 8.0.2
- **JDK:** java jdk version 8

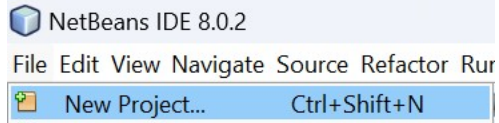
Downloads Required:

- Netbeans 8:
- Java JDK 8:

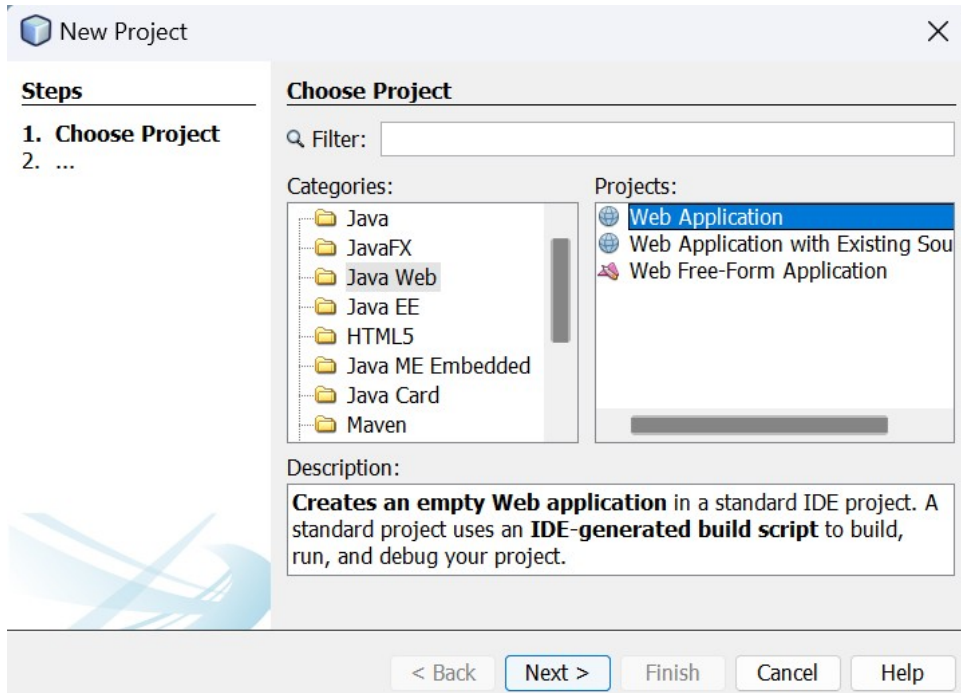
Project Name: Mtomapplication

Project Architecture:

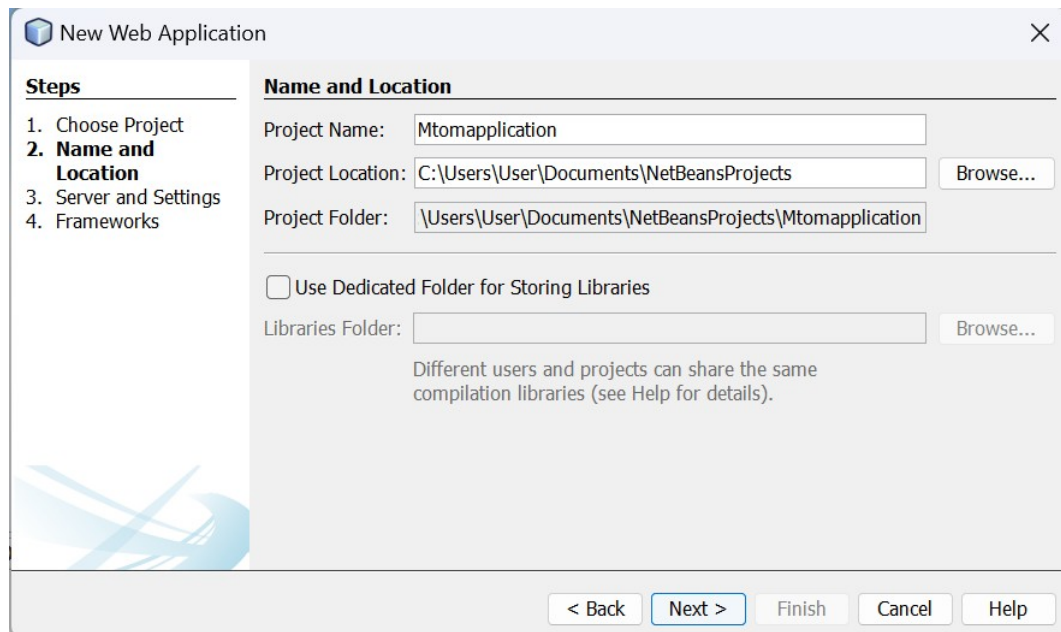




Select Java Web >Web Application



Click Next



New Web Application

Steps

1. Choose Project
- 2. Name and Location**
3. Server and Settings
4. Frameworks

Name and Location

Project Name:

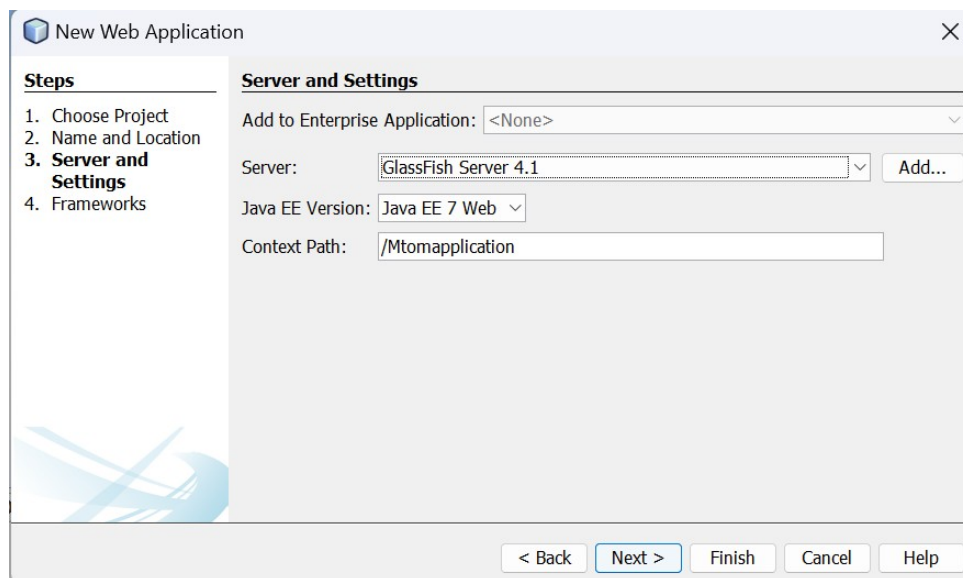
Project Location:

Project Folder:

☐ Use Dedicated Folder for Storing Libraries

Libraries Folder:

Different users and projects can share the same compilation libraries (see Help for details).



New Web Application

Steps

1. Choose Project
2. Name and Location
- 3. Server and Settings**
4. Frameworks

Server and Settings

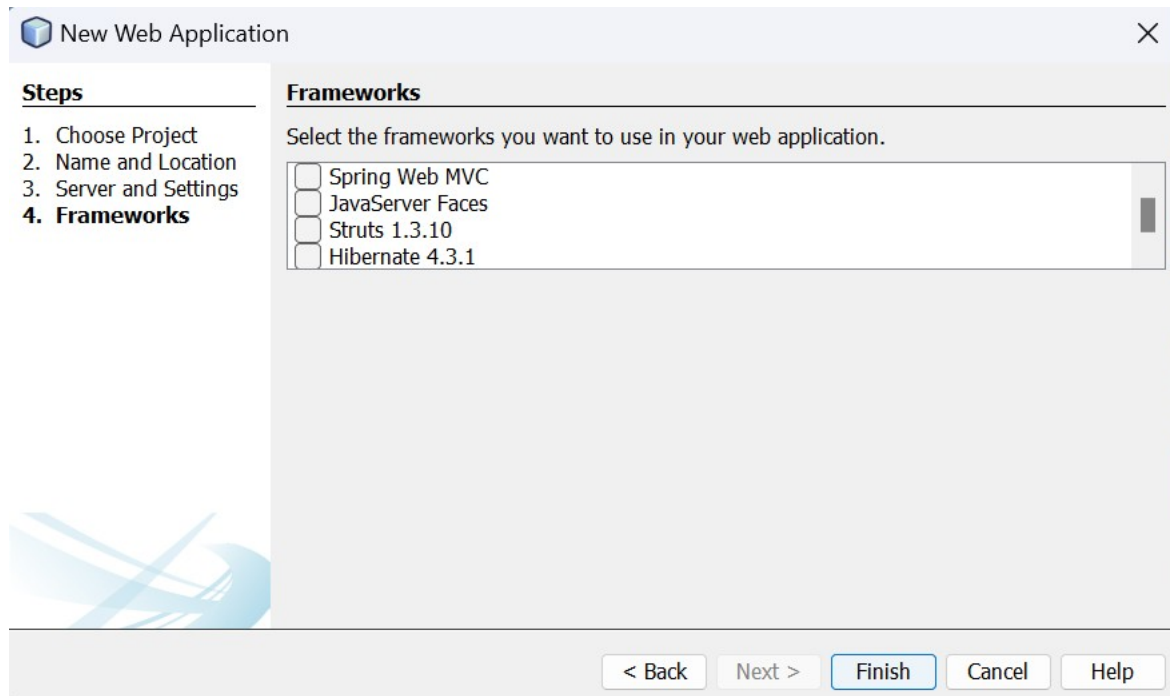
Add to Enterprise Application:

Server:

Java EE Version:

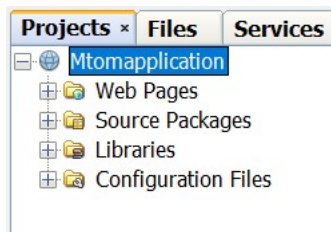
Context Path:

Click Next

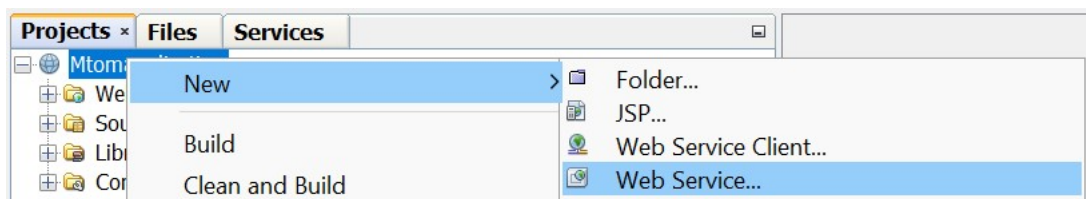


Click Finish

Expand Project Architecture



Right Click on Mtomapplication Project New > Web Service



Name: ImageWS and Package: mypkg and Finish

New Web Service

Steps

1. Choose File Type
2. **Name and Location**

Name and Location

Web Service Name:

Project:

Location:

Package:

☒ Create Web Service from Scratch

☐ Create Web Service from Existing Session Bean

Enterprise Bean:

☐ Implement Web Service as Stateless Session Bean

< Back Next > **Finish** Cancel Help

Filename: ImageWS.java

/*

* To change this license header, choose License Headers in Project Properties.

* To change this template file, choose Tools | Templates

* and open the template in the editor.

*/

```
package mypkg;
```

```
import java.io.*;
```

```
import javax.xml.ws.Oneway;
```

```
import javax.xml.ws.WebService;
```

```
import javax.xml.ws.WebMethod;
```

```
import javax.xml.ws.WebParam;
```

```
import javax.xml.ws.soap.MTOM;

@MTOM(enabled = true, threshold = 60000)
@WebService(serviceName = "ImageWS")
public class ImageWS {

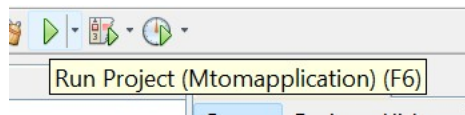
    @WebMethod(operationName = "upload")
    @Oneway
    public void upload(@WebParam(name = "Filename") String Filename, @WebParam(name =
"ImageBytes") byte[] ImageBytes) {
        String filePath = "C:/Picture/upload/" + Filename;
        try {
            FileOutputStream fos = new FileOutputStream(filePath);
            BufferedOutputStream bos = new BufferedOutputStream(fos);
            bos.write(ImageBytes);
            bos.close();
            System.out.println("Received file: " + filePath);
        } catch (Exception ex) {
            ex.printStackTrace();
        }
    }

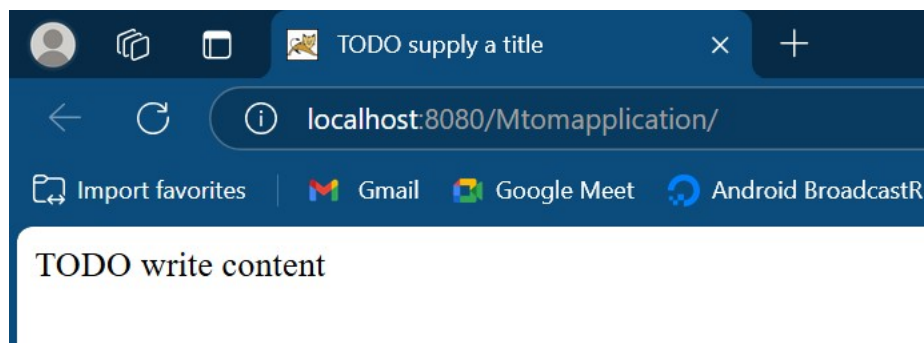
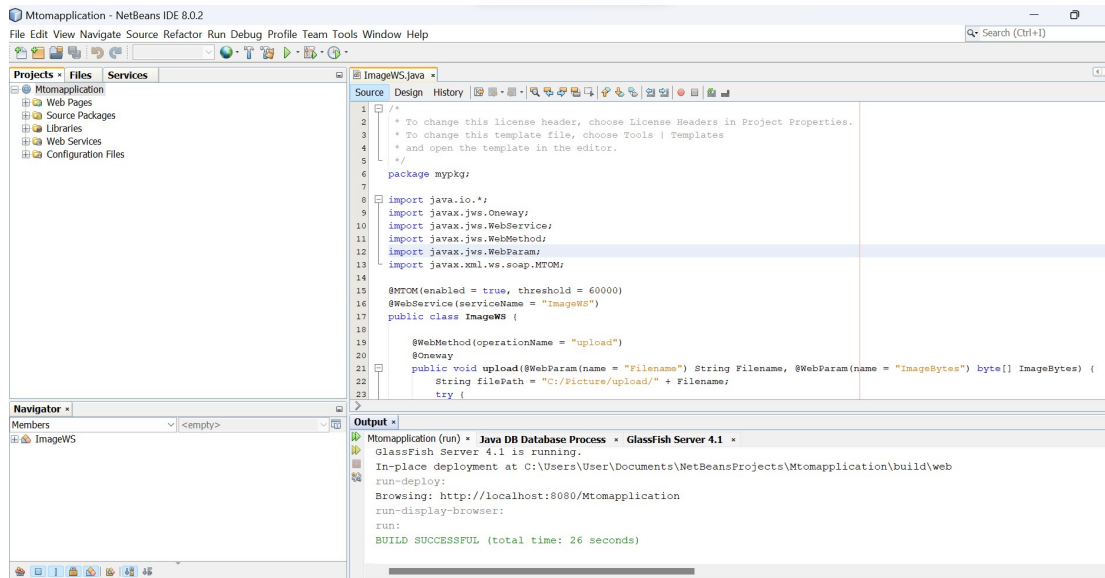
    @WebMethod(operationName = "download")
    public byte[] download(@WebParam(name = "Filename") String Filename) {
        String filePath = "C:/Picture/upload/" + Filename;
        System.out.println("Sending file: " + filePath);
        try {
```

```
File file = new File(filePath);  
FileInputStream fis = new FileInputStream(file);  
BufferedInputStream bis = new BufferedInputStream(fis);  
byte[] fileBytes = new byte[(int) file.length()];  
bis.read(fileBytes);  
bis.close();  
return fileBytes;  
} catch (Exception ex) {  
    ex.printStackTrace();  
    return null;  
}  
}  
}
```

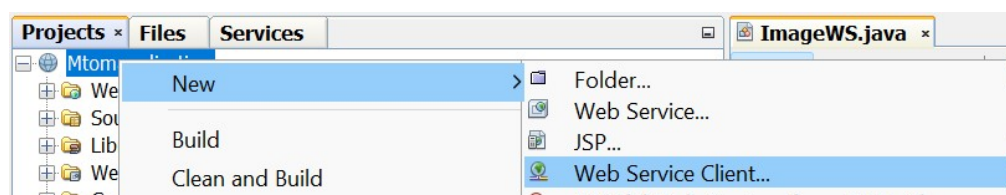
Save File.

Run Project

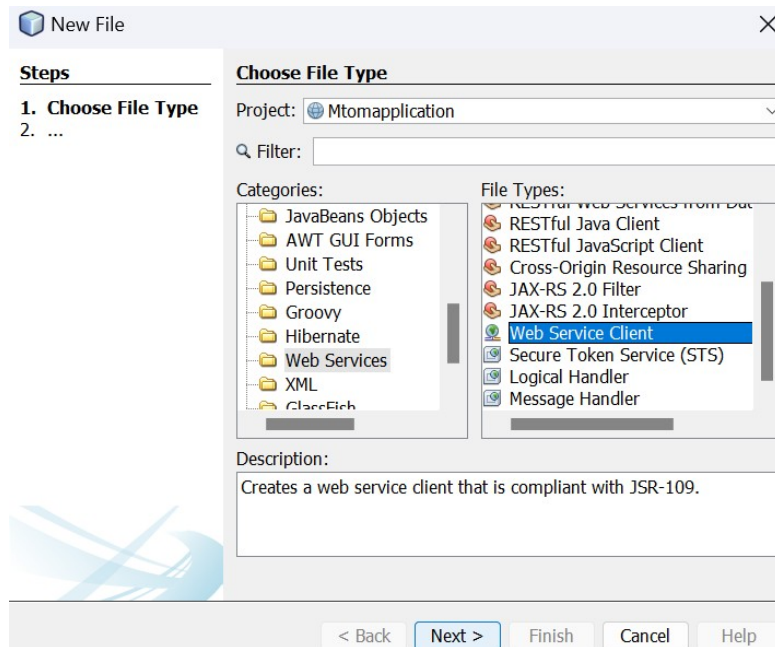




Right Click on Mtomapplication > New > Web Service Client.



If not available click on Others



Select on Browse

New Web Service Client

Steps

1. Choose File Type
2. **WSDL and Client Location**

WSDL and Client Location

Specify the WSDL file of the Web Service.

☒ Project: **Browse...**

☐ Local File: Browse...

☐ WSDL URL: Browse...

☐ IDE Registered: Browse...

Specify a package name where the client java artifacts will be generated:

Project: Mtomapplication

Package: pkg

☐ Generate Dispatch code

Enter the URL of the service you wish to use.

< Back Next > Finish Cancel Help

Browse Web Services

Web Services:

Mtomapplication

OK Cancel

Browse Web Services

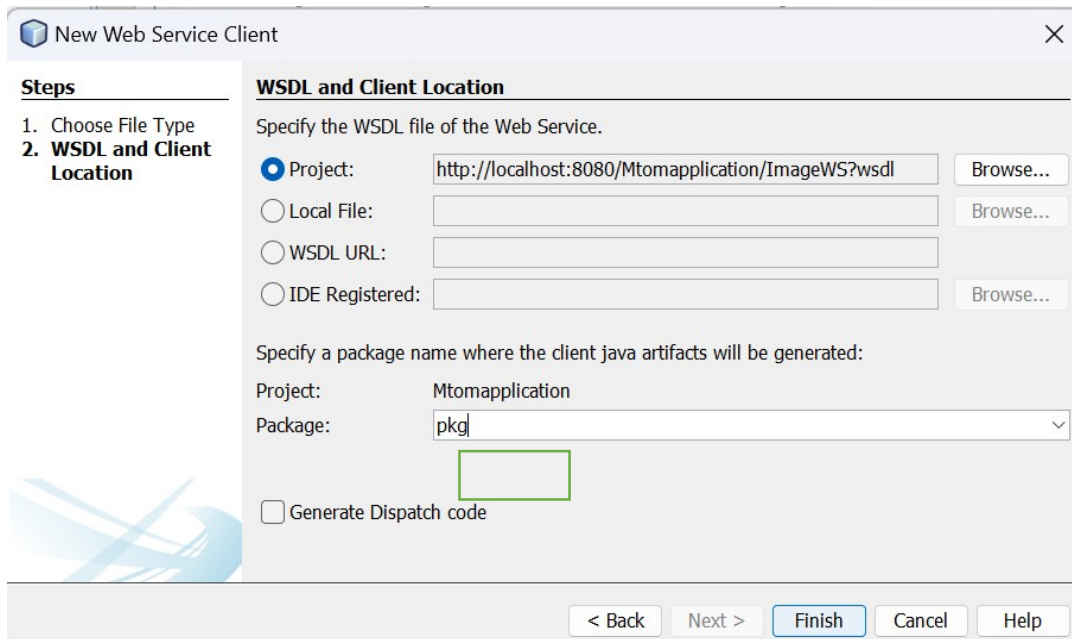
Web Services:

Mtomapplication

ImageWS

OK Cancel

Package: pkg

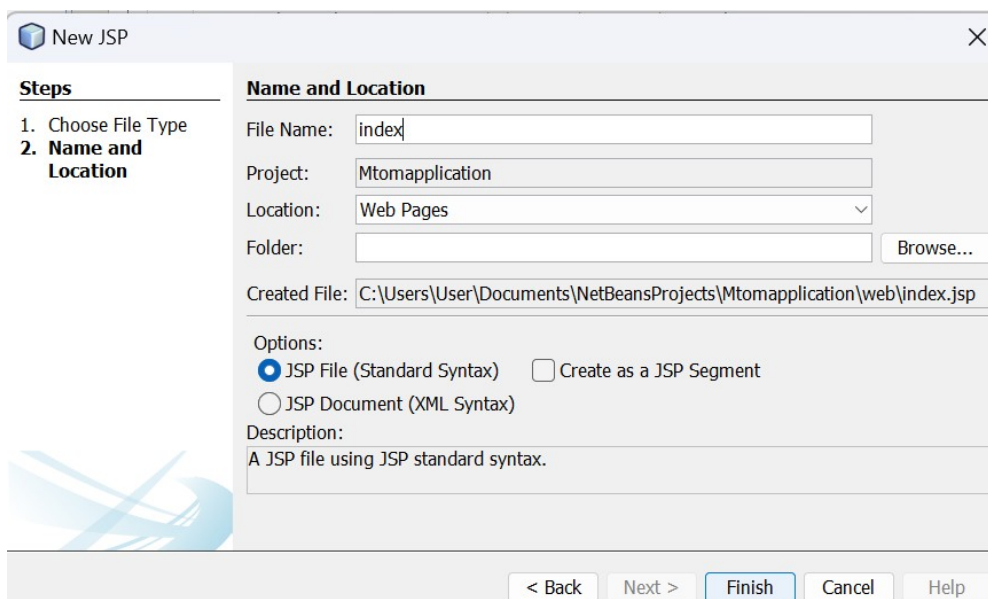


Finish

New > JSP



Name: index and click Finish



Filename: index.jsp

<%--

Document : index

Created on : Feb 3, 2025, 1:25:10 PM

Author : User

--%>

<%@page import="java.io.BufferedOutputStream"%>

<%@page import="java.io.FileOutputStream"%>

<%@page import="java.io.FileInputStream"%>

<%@page import="java.io.BufferedInputStream"%>

<%@page import="java.io.File"%>

<%@page import="javax.xml.ws.soap.MTOMFeature"%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>

<!DOCTYPE html>

<html>

<head>

<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">

<title>JSP Page</title>

</head>

<body>

<%-- start web service invocation --%><hr/>

<%

try {

pkg.ImageWS_Service service = new pkg.ImageWS_Service();

pkg.ImageWS port = service.getImageWSPort(new MTOMFeature(60000));

String filePath="C:/Picture/flower.jpg"; // Assuming this is the correct file path

File file=new File(filePath);

FileInputStream fis=new FileInputStream(file);

```

        BufferedInputStream bis=new BufferedInputStream(fis);
        String filename = file.getName();
        byte[]imageBytes=new byte[(int)file.length()];
        bis.read(imageBytes);
        port.upload(filename, imageBytes);
        bis.close();
        out.println("File uploaded :" + filePath);
    } catch (Exception ex) {
        // TODO handle custom exceptions here
        ex.printStackTrace();
    }
}
%>
<%-- end web service invocation --%><hr/>
<%-- start web service invocation --%><hr/>
<%
try {
    pkg.ImageWS_Service service = new pkg.ImageWS_Service();
    pkg.ImageWS port = service.getImageWSPort();
    // TODO initialize WS operation arguments here
    String filename = "flower.jpg";
    String filePath = "C:/Picture/download/" + filename;
    byte[] fileBytes = port.download(filename);
    // Write the downloaded bytes to a file
    FileOutputStream fos = new FileOutputStream(filePath);
    BufferedOutputStream bos = new BufferedOutputStream(fos);
    bos.write(fileBytes);
    bos.close();

```

```

        out.println("File downloaded: " + filePath);
    } catch (Exception ex) {
        // TODO handle custom exceptions here
        ex.printStackTrace();
    }
%>

<%-- end web service invocation --%><hr/>

</body>

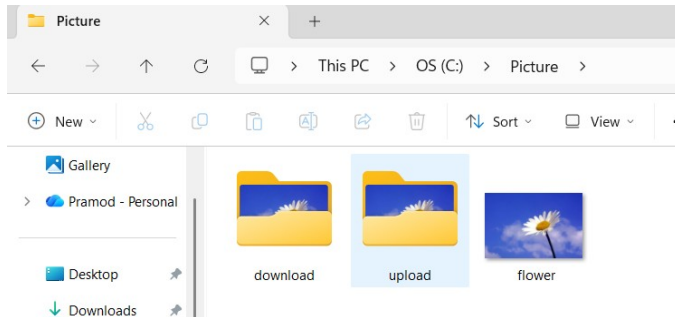
</html>

```

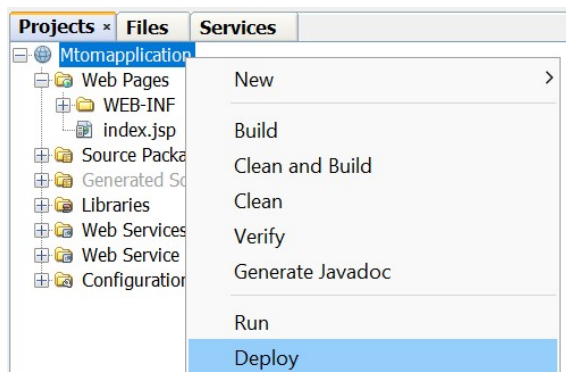
Create a Folder Name: Picture inside

Create 2 Folders name: download and upload and an image with name flower or any name with proper extension

Here image name and extension: flower.jpg



Deploy Project



Then Run Project or F6

